1.3	(7) $\Omega_2 = \frac{2}{3}\pi$	0.2	**0.7**
	(8) $dg_2 = \frac{dF}{m} = \alpha\Omega_2 = \frac{2}{3}\pi G\rho_2\Delta h$	0.2	
	(9) $g_2 = \frac{1}{3}\pi G\rho_2 a$	0.2	
	$g_2 = 3.14 m/s^2$	0.1	
1.4	(10) (11) $\frac{mv_1^2}{2} - U_1 = 0$ $\frac{mv_2^2}{2} - U_2 = 0$	0.2	**2.0**
	(12) $U_c = 6U_1\frac{\rho_3}{\rho_2}$	0.4	
	(13) $U = G\sum\frac{m\rho_3\Delta V_i}{r_i} \sim Gm\rho_3 a^2$	0.4	
	(14) $U_c = 8\frac{U_2}{4} = 2U_2$	0.2	
	(15) $U_2 = 3U_1\frac{\rho_3}{\rho_2}$	0.4	
	(16) $v_2 = \sqrt{\frac{3\rho_3}{\rho_2}}v_1$	0.3	
	$v_2 = 6{,}30 km/s$	0.1	
2.1	(17) $\frac{mv_\infty^2}{2} = \frac{mu_x^2}{2} + \frac{mu_y^2}{2} - G\frac{Mm}{R}$	0.2	**2.5**
	(18) $mu_x - mv_\infty = \int G\frac{Mm}{r^2}cos\varphi dt = \int G\frac{Mm}{r^2\dot{\varphi}}cos\varphi d\varphi$	0.4	
	(19) $r^2\dot{\varphi} = v_\infty b$	0.2	
	(20) $mu_x - mv_\infty = G\frac{Mm}{v_\infty b}sin\theta$	0.4	
	(21) $mu_y = G\frac{Mm}{v_\infty b}(1 - cos\theta)$	0.4	
	(23) $u_x = (1 + zsin\theta)v_\infty$	0.3	
	(24) $u_y = z(1 - cos\theta)v_\infty$	0.3	
	(26) (27) $\theta = \arcsin\frac{\frac{b}{R} - \frac{GM}{v_\infty^2 b}}{\sqrt{1+\left(\frac{GM}{v_\infty^2 b}\right)^2}} + \arcsin\frac{\frac{GM}{v_\infty^2 b}}{\sqrt{1+\left(\frac{GM}{v_\infty^2 b}\right)^2}}$ $\theta = 2\arctan\frac{1-\sqrt{1+2\frac{GM}{v_\infty^2 b}\frac{b}{R} - \frac{b^2}{R^2}}}{\frac{b}{R} - 2\frac{GM}{v_\infty^2 b}}$	0.2	
	$\theta = 0{,}789\, rad = 45{,}2°$	0.1	
2.2	(28) $F(r) = -G\frac{\rho_4\cdot\frac{4}{3}\pi r^3}{r^2}m = -\frac{4}{3}\pi G\rho_4 mr$	0.4	**2.0**
	(29) $U(r) = \frac{2}{3}\pi G\rho_4 mr^2 + C = G\frac{Mm}{2R^3}r^2 + C$	0.3	
	(30) $G\frac{Mm}{2R^3}R^2 + C = -G\frac{Mm}{R}$	0.4	
	(32) $\frac{mv_\infty^2}{2} = \frac{mv_0^2}{2} + \frac{GMm}{2R^3}r_{min}^2 - \frac{3GMm}{2R}$,	0.2	
	(33) $v_0 r_{min} = v_\infty b$	0.2	
	(35) $r_{min} = \sqrt{\frac{(\frac{3zb}{R}+1) \pm \sqrt{(\frac{3zb}{R}+1)^2 - \frac{4zb^3}{R^3}}}{2\frac{zb}{R^3}}}$	0.2	
	, (36)	0.2	
	$r_{min} = 4.97 \times 10^9 m$	0.1	
2.3	(37) $\frac{mv_{\infty,min}^2}{2} = \frac{m0^2}{2} + \frac{mu_\tau^2}{2} - G\frac{Mm}{R}$	0.4	**1.0**